Physics - Undergraduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. De Broglie hypothesized that the linear momentum and wavelength of a free massive particle are related by which of the following constants?
   (a) Planck's constant
   (b) Boltzmann's constant
   (c) The Rydberg constant
   (d) The speed of light

2. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
   (a) 2
   (b) 250
   (c) 5,000
   (d) 10,000

3. A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is
   (a) 4
   (b) 5
   (c) 6
   (d) 20

4. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
   (a) 2
   (b) 250
   (c) 5,000
   (d) 10,000

5. A rod measures 1.00 m in its rest system. How fast must an observer move parallel to the rod to measure its length to be 0.80 m?
   (a) 0.50 c
   (b) 0.60 c
   (c) 0.70 c
   (d) 0.80 c

True / False (10 marks)

Answer True or False

6. True or False: The emission spectrum of the doubly ionized lithium atom Li++ has wavelengths decreased by a factor of 9 compared to hydrogen.

7. True or False: The energy required to ionize a helium atom by removing one electron is 24.6 eV.

8. True or False: A meter stick moving at 0.8 c takes 1.6 ns to completely pass an observer in their reference frame.

9. True or False: The sign of the charge carriers in a doped semiconductor can be determined by measuring the Hall coefficient.

10. True or False: The total spin quantum number of the electrons in the ground state of neutral nitrogen (Z = 7) is 3/2.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the work done by an ideal monatomic gas during a quasi-static expansion to twice its volume under isothermal and adiabatic conditions. Analyze the relationship between the two work values.

12. Discuss the outcome of a one-dimensional, inelastic collision between a particle of mass 2 m and a stationary particle of mass m. Calculate and analyze the fraction of initial kinetic energy lost in the collision.

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